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github.com/aryashreep/learn-devops-in-90-days

DevOps & Cloud Training Series · 2026
Edition



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LINUX PERMISSIONS

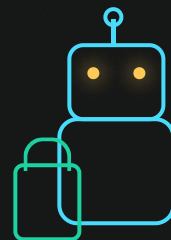
Day 03: File system ke "Security Guards" 



Golu

"Yaar Ladoo, ye **Permission Denied** wala error pichle 2 ghante se dimaag kharab kar raha hai!"

"Beep Boop! Golu, tension mat lo.
Aaj hum seekhenge ki Linux mein **Kaun,**
Kahan aur Kaise file access kar sakta hai."



Ladoo

Bhai, Aaj Kya Seekhenge?

*Aaj hum Linux ke "Security Guards" (Permissions) se dosti karenge.
Hum dekhenge ki kaise **4-2-1** ke maths se hum server ko secure bana sakte hain!*



LEARNING OBJECTIVES

→ **Understand:** UGO security model (Owner, Group, Others).

→ **Master:** rwx flags and their numeric (4-2-1) values.

→ **Standardize:** Use 755 for scripts and 644 for configs.

💡 **Kyu Seekhein?** Kyunki ek galat permission pure company ka server hack karwa sakti hai!



ANALOGY



Golu

"Bhai, Linux permissions ek **Secure Office Building** jaisa hai! Staff ke paas ID badge hota hai, par guest sirf reception tak ja sakta hai."

1. **Users:** Staff members (Malik/Golu)
2. **Groups:** Same department (Team members)
3. **Others:** Outsiders (Baaki duniya)

? DID YOU KNOW?

0

Root's UID

777

Security Disaster!

644

Default for Files



KEY DIRECTORIES — WHAT LIVES WHERE

-  **/etc** → All Configuration files
 -  **/bin** → Essential Commands (ls, cd, cp)
 -  **/var/log** → Log files (History book)
 -  **/home** → User's personal space
-



KEY CONCEPTS

- rwx r-x r--

7

5

4

Logic: $r(4) + w(2) + x(1)$



COMPARE (WHAT CAN GO WRONG)

Dangerous (777)

"Pure world ko ghar ki chabi de di. Anyone can delete your DB!"

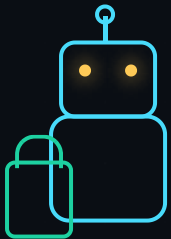
VS

Secure (644/755)

"Least Privilege: Jitni zaroorat hai, bas utna hi access."



REAL-WORLD EXAMPLE



Ladoo

"Beep Boop! Golu, meri website pe **403 Forbidden** aa raha hai! Nginx script read nahi kar paa raha."

Fix: `chown www-data:www-data index.html && chmod 644 index.html`

⚠ COMMON MISCONCEPTIONS

- **Myth:** "777" is the easiest way to fix any error.
- **Reality:** It's a huge risk. Always fix ownership first!
- **Myth:** Root needs permission to read files.
- **Reality:** Root is the Linux GOD. Usey koi nahi rok sakta!

▶ TRY IT YOURSELF

1. `touch secret.sh` *(Golu ki naya file)*
2. `chmod 400 secret.sh` *(Try to read it!)*
3. `chmod +x secret.sh` *(Ab power mode on!)*

✓ QUICK SUMMARY

- **UGO:** User, Group, Others.
 - **4-2-1:** Read, Write, Execute values.
 - **chmod:** Change Permissions.
 - **sudo:** Power of Root for one command.
-

? MINI QUIZ — TEST YOUR UNDERSTANDING

1. Read (r) ka numeric value kya hai?

- a) 1
- b) 2
- c) 4
- d) 7

2. Agar permission string "-rwxr-xr-x" hai, toh octal value kya hogi?

- a) 644
- b) 755
- c) 777
- d) 700

3. Nginx web server ke configuration files standardly kahan milti hain?

- a) /home
- b) /bin
- c) /etc/nginx
- d) /var/log

4. Ek bash script ko "Execute" permission dene ke liye kaunsa command use karenge?

- a) chmod +x
- b) chown +x
- c) sudo run
- d) chmod 444

5. "Root" user ka User ID (UID) kya hota hai?

- a) 1000
- b) 1
- c) 0
- d) 999

6. Agar aapne "chmod 000 file.txt" kar diya, toh kya Root usey padh payega?

- a) Nahi
- b) Haan
- c) Sirf sudo ke saath
- d) Error aayega

7. Production server par "chmod 777" karna kaisa hai?

a) Best practice

b) Quick fix

c) Security Disaster

d) Normal process

✓ ANSWER KEY

"Golu, pehle khud try kar, phir match kar!"

1-c | 2-b | 3-c | 4-a | 5-c | 6-b | 7-c

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Explore the Repo: github.com/aryashreep/learn-devops-in-90-days